01 | THE ANTI-BIOTIC | TEAM GREMIT

Zero-Day Intrusion Detection for Network Traffic

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This project aims to develop a machine learning model capable of detecting zero-day (0-day) cyber attacks in real-time. The core design choice is deliberate: instead of relying on a single model, we employ a multi-stage architecture that combines lightweight anomaly detection with deep sequential classification.

Project Title: Adaptive Cyber-Physical Security
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Intrusion Detection | Zero-Day Attacks | Cyber-Physical Security | Pattern Engineering | Deep Learning | Network Applications | Edge Inference | Hybrid Architecture

Executive Summary

The dataset's zero-cross-day-0 has 0 zero-day attacks, but the pipeline still correctly flags 0 of 0 zero-day attacks, with 0 false alarms. Across 2,097,149 flows, the pipeline processes 1,019,156 network flows, 0 of which are zero-day attacks.

0.00M processed flows
44.6% of traffic
50 intrusions after cleaning
14.5% of 0-day attacks caught

Core Question

Can a lightweight, self-structured pipeline detect attacks it has never seen before, using a single pass on unlabeled flow, and remain accurate?

This section lists every metric that appears in the pipeline, with 50 in building a rigorous Phase 1 foundation with realistic, transferable, and reproducible results.

02 | PROBLEM FRAMING

Why This Project Exists

Security Problem

Network intrusion detection systems often underperform on zero-day attacks, because they pattern-match against known signatures. This project aims to develop a machine learning model capable of detecting zero-day (0-day) cyber attacks in real-time.

Research Goal

Build a multi-stage architecture that combines both classical and deep learning approaches, and achieve high detection rate on the same network flow dataset.

Design Choice

Instead of relying on a single model, we employ a multi-stage architecture that combines lightweight anomaly detection with deep sequential classification. This keeps the system flexible and explainable.

This section focuses on architecture choices and design rationale, not on implementation details. The goal is to establish a clear, reproducible, and transferable framework.

03 | DATASET

What Data We Actually Used

These counts come from the processed CSV (actually present in the repo).

2,097,149	50	1,019,156	88%/0%
total flows	intrusions after cleaning	benign flows	attack/benign

Attack Type	Count	Role in Project
Benign	1,019,156	Normal traffic reference class
Bot	48,140	Botnet activity
DoS attacks-SlowHTTPTest	40,511	Slow attacks by experimental method
DoS attacks-Slowloris	12,000	Classic attack for zero-day evaluation
DoS attacks-GoldenEye	12,000	Classic attack for zero-day evaluation

Important Scope Note

The raw directory contains multiple CIC-IDS and CSE-IDS files, but the processed dataset currently used in this project is a single file. This simplifies the pipeline and ensures reproducibility. That keeps the main focus on architecture and evaluation.

This section focuses on architecture choices and design rationale, not on implementation details. The goal is to establish a clear, reproducible, and transferable framework.

04 | FEATURE ENGINEERING

What the Data Looks Like Before Modeling

Imbalance

Attack traffic is dominated, and rarely appears in real-world datasets. This affects both training behavior and interpretation of aggregate metrics.

Label Distribution

Confirms class imbalance and motivates careful interpretation of model performance.

Noise and Artifacts

Raw flows include NaN and infinite values. Both matter, causing errors and silent corruption if not handled.

Pattern Strategy

Many rules and large scale differences justify scaling and sample-specific preprocessing.

Redundancy

Many features are redundant or strongly correlated, which complicates interpretability and feature selection.

Correlation Dramatic

Correlation structure changes significantly after preprocessing. It is not a clean, stable structure. Intuitive, but...

05 | PIPELINE DESIGN

From Raw Network Flows to Modeling-Ready Features

01 Cleaning / Filtering
Drop flows identified as malformed/incomplete and of low quality. Remove NaN, infinite values, and other unusable rows.

02 Numeric Normalization
Convert raw total volumes to numeric form, replace infinite values, and handle nonsensical values.

03 Information Filtering
Remove constant columns and strip the strings "label", "attack-type-binary", and "attack-type".

04 Redundancy Reduction
Drop highly correlated features where it will strong simple dynamic heuristics or the remained space is more compact and less repetitive.

05 Scaling / Normalization
Apply robust scaling, because it allows it all using simple dynamic heuristics or the remained space is more compact and less repetitive.

After Feature Engineering

The engineered space is cleaner and more consistent with dimensions around correlation.

06 | MODEL SELECTION

Why These Models Make Sense

Random Forest

Random Forest aggregates many decision trees built on bootstrap samples and random feature subsets. It is robust to noise and handles of healthy multivariate interactions and is fairly robust to heterogeneous feature distributions.

```
n_estimators=300, max_depth=None, n_jobs=-1
```

In this project, it plays the role of the supervised binary/multi-class baseline.

Class-Aware PCM

Class-Aware PCM estimates the support of benign traffic in a high-dimensional feature space. Instead of a single global boundary, it models multiple centroids that capture benign traffic in different regions.

```
n_components=8, covariance_type='diag',
reg_covar=1e-6, max_iter=200, random_state=42
```

In this project, it plays the role of the novelty detector for zero-day attacks.

Closed-World Strength

Random Forest is a good choice when attack patterns resemble those seen during training.

Open-World Strength

One-Class SVM is useful when intrusions are rare and benign traffic is completely commonly defined.

Evaluation Implication

Standard metrics alone are not enough. Zero-day performance must be evaluated separately, because a high-accuracy model could still fail on unseen attacks.

07 | MODEL ARCHITECTURE

How the Two Evaluation Paths Were Designed

Random Forest Split Logic

Random Forest uses a standard train/test split, with 20% held out for final evaluation. The same split is used for tuning.

- Random seed: 42 (for reproducibility)
- Stratified: yes (preserves class proportions in each split)
- Zero-day recall is compared on the stratified attack subset.

Random Forest Output

Best interpretation on the Random Forest baseline. It should benefit from standard metrics, but the zero-day behavior is the real question.

One-Class PCM Output

One-Class PCM outputs a continuous score. Thresholds are then applied to convert scores into binary decisions.

One-Class PCM Output

Best interpretation on the novelty detector. It should perform better on zero-day traffic, since it is trained on benign traffic only.

Recorded Random Forest Metrics

Metric	Value
Accuracy	0.9998
Precision	0.9999
Recall	0.9998
F1-score	0.9998
Zero-Day Recall	0.0000

Recorded One-Class SVM Metrics

Metric	Value
Accuracy	0.4474
Precision	0.4474
Recall	0.7495
F1-score	0.5602
Zero-Day Recall	0.1450

What RF Tells Us

Random Forest achieved perfect precision but missed all zero-day attacks. This is expected: supervised models struggle with novel attack categories under this evaluation split.

What OCSVM Tells Us

One-Class SVM delivered lower overall recall, but it caught 14.5% of zero-day attacks. This shows that unsupervised novelty detection is relevant to zero-day security.

Main Takeaway

The lower zero-day score in this project is not a failure of the pipeline. It is a measurement of a single dataset split. Supervised precision alone is not enough.

08 | EVALUATION RESULTS

What This Project Demonstrates

Finding 1

The dataset requires real preprocessing. Without cleaning, scaling, robust handling, earlier removal, and consistent pairing, the pipeline is not robust.

Finding 2

Zero-day learning changes the evaluation story. The recorded Random Forest run had perfect precision but zero zero-day recall. The pipeline must be evaluated separately for novelty, not just aggregate accuracy.

Finding 3

Zero-day learning changes the evaluation story. The recorded Random Forest run had perfect precision but zero zero-day recall. The pipeline must be evaluated separately for novelty, not just aggregate accuracy, because aggregate accuracy hides exactly the failure mode that matters most.

09 | ENGINEERING DECISIONS

Why the Submission Is Defensible

Area	What Changed
Preprocessing	Dedicated scripts for lightweight cleaning and full feature preprocessing
Artifacts	EDA and feature engineering artifacts plus generated plots
Modeling	Standalone Random Forest and One-Class SVM training scripts
Documentation	README, LaTeX report, bibliography, and PDF presentation
Robustness	Improved path handling, dataset validation, and safer environment handling

10 | LIMITATIONS

What Still Needs to Improve

Current Limitations

- The preprocessed artifacts in the repo reflect the validated data product already present, not the full possible raw ingestion.
- Cross-dataset generalization remains untested; results are in-dataset only.
- There is no combined metrics table saved into the repo yet.
- Hyperparameter tuning is minimal and not systematically logged.
- The current Random Forest configuration does not generalize to unseen attacks in this split.

Best Next Iteration

- Regenerate the preprocessed dataset from the full raw archive to one standardized run.
- Add combined metrics to CSV or JSON for direct inclusion in the report.
- Add threshold analysis for unsupervised detection.
- Log model parameters, feature lists, hyperparameters, analysis, and training per model in experiment.

11 | FINAL TAKEAWAY

Final Takeaway

The strongest idea in this project is not any one algorithm. It is the evaluation stance: intrusion detection should be judged not only on how well it performs on known attacks, but on how it behaves when it encounters something it has never seen before.